

Berezin Transform

Brian Weiner

Penn State Univ

(Dated: 08/29/2010)

Abstract

Contents

I. Complex Integral Representations on Holomorphic Wavefunctions	2
A. Algebraic Structure	8
B. Independent Particle Example	9
C. BiOrthogonal Basis	11
II. Extra	11
A.	12

I. COMPLEX INTEGRAL REPRESENTATIONS ON HOLOMORPHIC WAVEFUNCTIONS

Using

$$I = \int |\mathbf{z}\rangle \langle \mathbf{z}| d\mu(\mathbf{z}) = \int \frac{|\mathbf{z}\rangle \langle \mathbf{z}|}{\langle \mathbf{z} | \mathbf{z} \rangle} d\tilde{\mu}(\mathbf{z}), \quad (1)$$

where

$$d\tilde{\mu}(\mathbf{z}) = \langle \mathbf{z} | \mathbf{z} \rangle d\mu(\mathbf{z}) \quad (2)$$

and

$$A = IAI \quad (3)$$

one can construct a biholomorphic representations of an operator A wrt to the measures μ and $\tilde{\mu}$ by

$$\begin{aligned}
A &= \int |\mathbf{z}_1\rangle \langle \mathbf{z}_1| A |\mathbf{z}_2\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\
&= \int \frac{|\mathbf{z}_1\rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle}} \frac{\langle \mathbf{z}_1 | A \mathbf{z}_2 \rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} \frac{\langle \mathbf{z}_2 |}{\sqrt{\langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} d\tilde{\mu}(\mathbf{z}_1) d\tilde{\mu}(\mathbf{z}_2) \\
&= \int Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) |\mathbf{z}_1\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\
&= \int \tilde{Q}_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) \frac{|\mathbf{z}_1\rangle \langle \mathbf{z}_2|}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} d\tilde{\mu}(\mathbf{z}_1) d\tilde{\mu}(\mathbf{z}_2), \quad (4)
\end{aligned}$$

where

$$\begin{aligned}
Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) &= \langle \mathbf{z}_1 | A \mathbf{z}_2 \rangle \\
\tilde{Q}_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) &= \frac{\langle \mathbf{z}_1 | A \mathbf{z}_2 \rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}}, \quad (5a)
\end{aligned}$$

that have the following action on a holomorphic wavefunction

$$\begin{aligned}
A|g\rangle &= |Ag\rangle = \\
\int |\mathbf{z}_1\rangle \langle \mathbf{z}_1| A |\mathbf{z}_2\rangle \langle \mathbf{z}_2| |g\rangle d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) &= \int Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) |\mathbf{z}_1\rangle \langle \mathbf{z}_2| |g\rangle d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\
&= \int (Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) \langle \mathbf{z}_2| |g\rangle d\mu(\mathbf{z}_2)) |\mathbf{z}_1\rangle d\mu(\mathbf{z}_1) \\
&= \int (Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) g(\bar{\mathbf{z}}_2) d\mu(\mathbf{z}_2)) |\mathbf{z}_1\rangle d\mu(\mathbf{z}_1) \\
[Ag](\bar{\mathbf{z}}_1) &= \int (Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) g(\bar{\mathbf{z}}_2) d\mu(\mathbf{z}_2))
\end{aligned} \tag{6}$$

and

$$\begin{aligned}
A|g\rangle &= \int \frac{|\mathbf{z}_1\rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle}} \frac{\langle \mathbf{z}_1 | A \mathbf{z}_2 \rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} \frac{\langle \mathbf{z}_2 |}{\sqrt{\langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} |g\rangle d\tilde{\mu}(\mathbf{z}_1) d\tilde{\mu}(\mathbf{z}_2) \\
&= \int \left(\tilde{Q}_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) \tilde{g}(\bar{\mathbf{z}}_2) d\tilde{\mu}(\mathbf{z}_2) \right) \frac{|\mathbf{z}_1\rangle}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle}} d\tilde{\mu}(\mathbf{z}_1) \\
\widetilde{[Ag]}(\bar{\mathbf{z}}_1) &= \int \tilde{Q}_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) \tilde{g}(\bar{\mathbf{z}}_2) d\tilde{\mu}(\mathbf{z}_2).
\end{aligned} \tag{7}$$

The ket-bra expansion of an operator A in the reference basis $\{|\varphi_{j_1}\varphi_{j_2}\rangle \langle \varphi_{k_1}\varphi_{k_2}| \mid 1 \leq j_1 < j_2, k_1 < k_2 \leq r\}$ is

$$\begin{aligned}
A &= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} |\varphi_{j_1}\varphi_{j_2}\rangle \langle \varphi_{j_1}\varphi_{j_2}| A |\varphi_{k_1}\varphi_{k_2}\rangle \langle \varphi_{k_1}\varphi_{k_2}| \\
&= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1 j_2 k_1 k_2} |\varphi_{j_1}\varphi_{j_2}\rangle \langle \varphi_{k_1}\varphi_{k_2}|,
\end{aligned} \tag{8}$$

where

$$A_{j_1 j_2 k_1 k_2} = \text{Tr} \{ |\varphi_{k_1}\varphi_{k_2}\rangle \langle \varphi_{j_1}\varphi_{j_2}| A \} = \text{Tr} \left\{ |\varphi_{j_1}\varphi_{j_2}\rangle \langle \varphi_{k_1}\varphi_{k_2}|^\dagger A \right\}. \tag{9}$$

The biholomorphic integral kernel Q_A can thus be expanded as

$$\begin{aligned}
Q_A(\bar{\mathbf{z}}_1, \mathbf{z}_2) &= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1 j_2 k_1 k_2} \langle \mathbf{z}_1 | \varphi_{j_1}\varphi_{j_2} \rangle \langle \varphi_{k_1}\varphi_{k_2} | \mathbf{z}_2 \rangle \\
&= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1 j_2 k_1 k_2} Q_{k_1 k_2 j_1 j_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) \\
&= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1 j_2 k_1 k_2} \tilde{Q}_{k_1 k_2 j_1 j_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) \sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle},
\end{aligned} \tag{10}$$

where the biholomorphic integral kernels of the bases are related by

$$Q_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) = \tilde{Q}_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) \sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle} \tag{11}$$

and

$$\begin{aligned}\tilde{Q}_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) &= \frac{\text{Tr} \{ |\varphi_{k_1} \varphi_{k_2} \rangle \langle \varphi_{j_1} \varphi_{j_2} | | \mathbf{z}_2 \rangle \langle \mathbf{z}_1 | \}}{\sqrt{\langle \mathbf{z}_1 | \mathbf{z}_1 \rangle \langle \mathbf{z}_2 | \mathbf{z}_2 \rangle}} \\ Q_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) &= \text{Tr} \{ |\varphi_{k_1} \varphi_{k_2} \rangle \langle \varphi_{j_1} \varphi_{j_2} | | \mathbf{z}_2 \rangle \langle \mathbf{z}_1 | \}. \end{aligned} \quad (12a)$$

The kernels $\{Q_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) | 1 \leq j_1 < j_2, k_1 < k_2 \leq r\}$ form a biholomorphic conb for $\mathfrak{F}_{op} = \mathcal{R}_{hol}(\mathcal{B}_2(\mathcal{H}^2)) \subset L_2(\mathbb{C}^{4(r-2)}, \mu)$ as

$$\begin{aligned} & \int \overline{Q_{j_1 j_2 k_1 k_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2)} Q_{p_1 p_2 q_1 q_2}(\bar{\mathbf{z}}_1, \mathbf{z}_2) d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\ &= \int \overline{\langle \mathbf{z}_1 | \varphi_{j_1} \varphi_{j_2} \rangle \langle \varphi_{k_1} \varphi_{k_2} | \mathbf{z}_2 \rangle} \langle \mathbf{z}_1 | \varphi_{p_1} \varphi_{p_2} \rangle \langle \varphi_{q_1} \varphi_{q_2} | \mathbf{z}_2 \rangle d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\ &= \int \langle \varphi_{q_1} \varphi_{q_2} | \mathbf{z}_2 \rangle \langle \mathbf{z}_2 | \varphi_{k_1} \varphi_{k_2} \rangle \langle \varphi_{j_1} \varphi_{j_2} | \mathbf{z}_1 \rangle \langle \mathbf{z}_1 | \varphi_{p_1} \varphi_{p_2} \rangle d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \\ &= \langle \varphi_{q_1} \varphi_{q_2} | \varphi_{k_1} \varphi_{k_2} \rangle \langle \varphi_{j_1} \varphi_{j_2} | \varphi_{p_1} \varphi_{p_2} \rangle = \delta_{(q_1 q_2)(k_1 k_2)} \delta_{(j_1 j_2)(p_1 p_2)}. \end{aligned} \quad (13)$$

The Q-symbols, $\{Q_{j_1 j_2 k_1 k_2}(\mathbf{z})\}$ and $\{\tilde{Q}_{j_1 j_2 k_1 k_2}\}$, wrt the measures μ and $\tilde{\mu}$ of the ket-bra basis elements are given by

$$Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) = \text{Tr} \{ |\varphi_{k_1} \varphi_{k_2} \rangle \langle \varphi_{j_1} \varphi_{j_2} | | \mathbf{z} \rangle \langle \mathbf{z} | \} = \tilde{Q}_{j_1 j_2 k_1 k_2}(\mathbf{z}) \langle \mathbf{z} | \mathbf{z} \rangle \quad (14)$$

and thus the Q-symbol, $Q_A(\mathbf{z})$ and $\tilde{Q}_A(\mathbf{z})$, of the operator A by

$$Q_A(\mathbf{z}) = \tilde{Q}_A(\mathbf{z}) \langle \mathbf{z} | \mathbf{z} \rangle = \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1 j_2 k_1 k_2} \tilde{Q}_{j_1 j_2 k_1 k_2}(\mathbf{z}) \langle \mathbf{z} | \mathbf{z} \rangle. \quad (15)$$

The set $\{Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) | 1 \leq j_1 < j_2, k_1 < k_2 \leq r\}$ is linearly independent and forms a basis for the space, $\mathfrak{F} \subset L_2(\mathbb{C}^{2(r-2)}, \mu)$, of Q-symbols, whose dimension is $\binom{r}{2}^2$. However, in general these functions are not holomorphic, biholomorphic, normalized nor orthogonal and their metric matrix is given by

$$\begin{aligned} \Delta_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} &= (Q_{l_1 l_2 m_1 m_2} | Q_{j_1 j_2 k_1 k_2})_{\mathfrak{F}} = \\ &= \int \overline{\text{Tr} \{ | \mathbf{z} \rangle \langle \mathbf{z} | | \varphi_{l_1} \varphi_{l_2} \rangle \langle \varphi_{m_1} \varphi_{m_2} | \}} \text{Tr} \{ | \mathbf{z} \rangle \langle \mathbf{z} | | \varphi_{j_1} \varphi_{j_2} \rangle \langle \varphi_{k_1} \varphi_{k_2} | \} d\mu(\mathbf{z}) \\ &= \int_{\mathbb{C}^{2(r-2)}} \overline{\gamma_2(\mathbf{z})}_{m_1 m_2 l_1 l_2} \gamma_2(\mathbf{z})_{j_1 j_2 k_1 k_2} d\mu(\mathbf{z}). \end{aligned} \quad (16)$$

One can form a (in general non unique) biorthogonal basis $\{\mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) | 1 \leq j_1 < j_2, k_1 < k_2 \leq r\}$ to $\{Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) | 1 \leq j_1 < j_2, k_1 < k_2 \leq r\}$ that

also spans $\mathfrak{F}$, i.e.

$$\begin{aligned} (\mathcal{P}_{j_1 j_2 k_1 k_2} | Q_{l_1 l_2 m_1 m_2})_{\mathfrak{F}} &= \int \overline{\mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z})} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) d\mu(\mathbf{z}) \\ &= \delta_{(l_1 l_2)(j_1 j_2)} \delta_{(m_1 m_2)(k_1 k_2)}. \end{aligned} \quad (17)$$

The $\{\mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z})\}$ basis can be expanded in terms of the $\{Q_{j_1 j_2 k_1 k_2}(\mathbf{z})\}$ basis as

$$\begin{aligned} \mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) &= \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} Q_{p_1 p_2 q_1 q_2}(\mathbf{z}) \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1} \\ &= \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} \gamma_2(\mathbf{z})_{p_1 p_2 q_1 q_2} \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1}, \end{aligned} \quad (18)$$

i.e.

$$|\mathcal{P}\rangle = |Q\rangle \Delta^{-1}. \quad (19)$$

This leads to

$$(\mathcal{P} | \mathcal{P}) = \Delta^{-1} (Q | Q) \Delta^{-1} = \Delta^{-1}, \quad (20)$$

$$\begin{aligned} |Q\rangle &= |\mathcal{P}\rangle \Delta, \\ |\mathcal{P}\rangle &= |Q\rangle \Delta^{-1}, \end{aligned} \quad (21)$$

and

$$I_{\mathfrak{F}} = |Q\rangle \Delta^{-1} (Q | = |\mathcal{P}\rangle \Delta (\mathcal{P} | = |\mathcal{P}\rangle (Q | = |Q\rangle (\mathcal{P} |. \quad (22)$$

One can also observe that

$$I_{\mathfrak{F}} = \int ||\mathbf{z}\rangle \langle \mathbf{z}|| (|\mathbf{z}\rangle \langle \mathbf{z}|) d\mu(\mathbf{z}), \quad (23)$$

which comes directly from

$$\begin{aligned} \Delta_{l_1 l_2 m_1 m_2 j_1 j_2 k_1 k_2} &= (Q_{l_1 l_2 m_1 m_2} | Q_{j_1 j_2 k_1 k_2}) = \\ &= \int_{\mathbb{C}^{2(r-2)}} \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}| \} \text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}| \} d\mu(\mathbf{z}) \\ &= \int_{\mathbb{C}^{2(r-2)}} (Q_{l_1 l_2 m_1 m_2} | |\mathbf{z}\rangle \langle \mathbf{z}|) (|\mathbf{z}\rangle \langle \mathbf{z}| | Q_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}). \end{aligned} \quad (24)$$

The orthogonality condition can be reexpressed as

$$\begin{aligned} \delta_{(l_1 l_2)(j_1 j_2)} \delta_{(m_1 m_2)(k_1 k_2)} &= \int \mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) Q_{l_1 l_2 m_1 m_2}(\mathbf{z}) d\mu(\mathbf{z}) \\ &= \int \mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) \overline{\text{Tr} \{ |\mathbf{z}\rangle \langle \mathbf{z}| |\varphi_{l_1} \varphi_{l_2}\rangle \langle \varphi_{m_1} \varphi_{m_2}| \}} d\mu(\mathbf{z}) \\ &= \text{Tr} \left\{ \left(\int |\mathbf{z}\rangle \langle \mathbf{z}| \mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) d\mu(\mathbf{z}) \right) |\varphi_{m_1} \varphi_{m_2}\rangle \langle \varphi_{l_1} \varphi_{l_2}| \right\}, \end{aligned} \quad (25)$$

which shows that the integral can be interpreted as the following operator

$$|\varphi_{j_1}\varphi_{j_2}\rangle\langle\varphi_{k_1}\varphi_{k_2}| \equiv \int \mathcal{P}_{j_1j_2k_1k_2}(\mathbf{z})|\mathbf{z}\rangle\langle\mathbf{z}|d\mu(\mathbf{z}) \quad (26)$$

and $\mathcal{P}_{j_1j_2k_1k_2}(\mathbf{z})$ as its P-symbol , i.e

$$P_{j_1j_2k_1k_2}(\mathbf{z}) = \mathcal{P}_{j_1j_2k_1k_2}(\mathbf{z}) , \quad (27)$$

note that P-symbols are not unique.

The P-symbol, $P_A(\mathbf{z})$, of any operator A can then be expressed as

$$P_A(\mathbf{z}) = \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} A_{j_1j_2k_1k_2} P_{j_1j_2k_1k_2}(\mathbf{z}) , \quad (28)$$

in particular

$$\begin{aligned} I_{\mathcal{H}^2} &= \sum_{1 \leq j_1 < j_2 \leq r} |\varphi_{j_1}\varphi_{j_2}\rangle\langle\varphi_{j_1}\varphi_{j_2}| = \sum_{1 \leq j_1 < j_2 \leq r} \int P_{j_1j_2j_1j_2}(\mathbf{z})|\mathbf{z}\rangle\langle\mathbf{z}|d\mu(\mathbf{z}) \\ &= \int \sum_{1 \leq j_1 < j_2 \leq r} P_{j_1j_2j_1j_2}(\mathbf{z})|\mathbf{z}\rangle\langle\mathbf{z}|d\mu(\mathbf{z}) = \int P_{I_{\mathcal{H}^2}}(\mathbf{z})|\mathbf{z}\rangle\langle\mathbf{z}|d\mu(\mathbf{z}) , \end{aligned} \quad (29)$$

thus

$$P_{I_{\mathcal{H}^2}}(\mathbf{z}) = \sum_{1 \leq j_1 < j_2 \leq r} P_{j_1j_2j_1j_2}(\mathbf{z}) = 1. \quad (30)$$

This gives

$$\langle\mathbf{z}''|\mathbf{z}'\rangle = \int \langle\mathbf{z}''|\mathbf{z}\rangle\langle\mathbf{z}|\mathbf{z}'\rangle d\mu(\mathbf{z}) \quad (31)$$

and

$$Q_{I_{\mathcal{H}^2}}(\mathbf{z}') = \int |\langle\mathbf{z}'|\mathbf{z}\rangle|^2 d\mu(\mathbf{z}) . \quad (32)$$

The reproducing kernel for $\mathcal{B}_2(\mathfrak{F})$ is

$$\begin{aligned} M(\mathbf{z}, \mathbf{z}') &= \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} (|\mathbf{z}\rangle\langle\mathbf{z}|P_{j_1j_2k_1k_2})(Q_{j_1j_2k_1k_2}|\mathbf{z}'\rangle\langle\mathbf{z}'|) \\ &= \sum_{\substack{1 \leq j_1 < j_2, k_1 < k_2 \leq r \\ 1 \leq p_1 < p_2, q_1 < q_2 \leq r}} (|\mathbf{z}\rangle\langle\mathbf{z}|Q_{j_1j_2k_1k_2})\Delta_{j_1j_2k_1k_2p_1p_2q_1q_2}^{-1}(Q_{p_1p_2q_1q_2}|\mathbf{z}'\rangle\langle\mathbf{z}'|) \\ &= \sum_{\substack{1 \leq j_1 < j_2, k_1 < k_2 \leq r \\ 1 \leq p_1 < p_2, q_1 < q_2 \leq r}} (|\mathbf{z}\rangle\langle\mathbf{z}|P_{j_1j_2k_1k_2})\Delta_{j_1j_2k_1k_2p_1p_2q_1q_2}(P_{p_1p_2q_1q_2}|\mathbf{z}'\rangle\langle\mathbf{z}'|) . \end{aligned} \quad (33)$$

The Kernel of the Berezin Transformation that transform P-symbols to Q-symbols

$$\mathfrak{B}\mathfrak{e} : \mathfrak{F} \rightarrow \mathfrak{F} \quad (34)$$

can be obtained by using the identity

$$|Q_{j_1 j_2 k_1 k_2}\rangle = \mathfrak{B}\mathfrak{e}(P_{j_1 j_2 k_1 k_2}) = \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} |Q_{p_1 p_2 q_1 q_2}\rangle (Q_{p_1 p_2 q_1 q_2} | P_{j_1 j_2 k_1 k_2}),$$

which leads to

$$\begin{aligned} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) &= [\mathfrak{B}\mathfrak{e}(P_{j_1 j_2 k_1 k_2})](\mathbf{z}) = \\ &= \int \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} (|\mathbf{z}\rangle \langle \mathbf{z}| | Q_{p_1 p_2 q_1 q_2}) (Q_{p_1 p_2 q_1 q_2} | |\mathbf{z}'\rangle \langle \mathbf{z}'|) (|\mathbf{z}'\rangle \langle \mathbf{z}'| | P_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}') \\ &= \int \mathfrak{B}\mathfrak{e}(\mathbf{z}, \mathbf{z}') (|\mathbf{z}'\rangle \langle \mathbf{z}'| | P_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}'). \end{aligned} \quad (35)$$

Thus

$$\mathfrak{B}\mathfrak{e}(\mathbf{z}, \mathbf{z}') = \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} (|\mathbf{z}\rangle \langle \mathbf{z}| | Q_{p_1 p_2 q_1 q_2}) (Q_{p_1 p_2 q_1 q_2} | |\mathbf{z}'\rangle \langle \mathbf{z}'|) = |\langle \mathbf{z} | \mathbf{z}' \rangle|^2. \quad (36)$$

The Kernel of the inverse Berezin Transformation that transform P-symbols to Q-symbols

$$\mathfrak{B}\mathfrak{e}^{-1} : \mathfrak{F} \rightarrow \mathfrak{F} \quad (37)$$

can be obtained by using the identity

$$|P_{j_1 j_2 k_1 k_2}\rangle = \mathfrak{B}\mathfrak{e}^{-1}(Q_{j_1 j_2 k_1 k_2}) = \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} |P_{p_1 p_2 q_1 q_2}\rangle (P_{p_1 p_2 q_1 q_2} | Q_{j_1 j_2 k_1 k_2}) \quad (38)$$

to give

$$\begin{aligned} P_{j_1 j_2 k_1 k_2}(\mathbf{z}) &= [\mathfrak{B}\mathfrak{e}^{-1}(Q_{j_1 j_2 k_1 k_2})](\mathbf{z}) = \\ &= \int \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} (|\mathbf{z}\rangle \langle \mathbf{z}| | P_{p_1 p_2 q_1 q_2}) (P_{p_1 p_2 q_1 q_2} | |\mathbf{z}'\rangle \langle \mathbf{z}'|) (|\mathbf{z}'\rangle \langle \mathbf{z}'| | Q_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}') \\ &= \int \sum_{\substack{1 \leq j_1 < j_2, k_1 < k_2 \leq r \\ 1 \leq p_1 < p_2, q_1 < q_2 \leq r}} (|\mathbf{z}\rangle \langle \mathbf{z}| | Q_{j_1 j_2 k_1 k_2}) \Delta_{j_1 j_2 k_1 k_2 p_1 p_2 q_1 q_2}^{-2} (Q_{p_1 p_2 q_1 q_2} | |\mathbf{z}'\rangle \langle \mathbf{z}'|) \\ &\quad \times (|\mathbf{z}'\rangle \langle \mathbf{z}'| | Q_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}') \\ &= \int \mathfrak{B}\mathfrak{e}^{-1}(\mathbf{z}, \mathbf{z}') (|\mathbf{z}'\rangle \langle \mathbf{z}'| | Q_{j_1 j_2 k_1 k_2}) d\mu(\mathbf{z}'), \end{aligned} \quad (39)$$

where

$$\mathfrak{B}\mathfrak{e}^{-1}(\mathbf{z}, \mathbf{z}') = \sum_{\substack{1 \leq j_1 < j_2, k_1 < k_2 \leq r \\ 1 \leq p_1 < p_2, q_1 < q_2 \leq r}} (|\mathbf{z}\rangle \langle \mathbf{z}| | Q_{j_1 j_2 k_1 k_2}) \Delta_{j_1 j_2 k_1 k_2 p_1 p_2 q_1 q_2}^{-2} (Q_{p_1 p_2 q_1 q_2} | |\mathbf{z}'\rangle \langle \mathbf{z}'|). \quad (40)$$

「The Berezin Transform gives

$$Q_{p_1 p_2 q_1 q_2}(\mathbf{z}) = \int P_{p_1 p_2 q_1 q_2}(\mathbf{z}') |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}'), \quad (41)$$

which leads to (using the biorthogonality relationships)

$$\begin{aligned} \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} P_{j_1 j_2 k_1 k_2}(\mathbf{z}) \Delta_{j_1 j_2 k_1 k_2 p_1 p_2 q_1 q_2} &= \int P_{p_1 p_2 q_1 q_2}(\mathbf{z}') |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \\ &\Downarrow \\ P_{l_1 l_2 m_1 m_2}(\mathbf{z}) &= \int \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}') \Delta_{j_1 j_2 k_1 k_2 l_1 l_2 m_1 m_2}^{-2} |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \end{aligned} \quad (42)$$

and

$$Q_{p_1 p_2 q_1 q_2}(\mathbf{z}) = \int \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}') \Delta_{j_1 j_2 k_1 k_2 p_1 p_2 q_1 q_2}^{-1} |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}'). \quad (43)$$

」

A. Algebraic Structure

a. *P-Symbol products*

$$\int \mathcal{P}_A(\mathbf{z}_1) |\mathbf{z}_1\rangle \langle \mathbf{z}_1| d\mu(\mathbf{z}_1) \int \mathcal{P}_A(\mathbf{z}_2) |\mathbf{z}_2\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_2) \quad (44)$$

$$= \int \mathcal{P}_A(\mathbf{z}_1) \mathcal{P}_B(\mathbf{z}_2) S(\bar{\mathbf{z}}_1, \mathbf{z}_2) |\mathbf{z}_1\rangle \langle \mathbf{z}_2| d\mu(\mathbf{z}_1) d\mu(\mathbf{z}_2) \quad (45)$$

$$= \int \mathcal{P}_{AB}(\mathbf{z}_1) |\mathbf{z}_1\rangle \langle \mathbf{z}_1| d\mu(\mathbf{z}_1) \quad (46)$$

$$\mathcal{P}_{AB}(\mathbf{z}_1) = \int \mathcal{P}_A(\mathbf{z}_1) S(\bar{\mathbf{z}}_1, \mathbf{z}_2) \mathcal{P}_B(\mathbf{z}_2) \quad (47)$$

b. *Q-Symbol products* Pointwise product produces a commutative w^* - algebra

$$Q_{A \cdot B}(\bar{\mathbf{z}}, \mathbf{z}) = Q_A(\bar{\mathbf{z}}, \mathbf{z}) Q_B(\bar{\mathbf{z}}, \mathbf{z}) \quad (48)$$

this product is distinct from non commutative products such as

$$Q_{A \circ B}(\bar{\mathbf{z}}, \mathbf{z}) = \frac{\langle \mathbf{z} | AB \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle}, \quad (49)$$

$$Q_{AB+}(\bar{\mathbf{z}}, \mathbf{z}) = \frac{1}{2} \frac{\langle \mathbf{z} | (AB + BA) \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \frac{1}{2} (Q_{A \circ B}(\bar{\mathbf{z}}, \mathbf{z}) + Q_{B \circ A}(\bar{\mathbf{z}}, \mathbf{z})). \quad (50)$$

B. Independent Particle Example

Let the ground state of a non interacting two particle system be

$$D_{IPS}^2 = |\varphi_1 \varphi_2\rangle \langle \varphi_1 \varphi_2| \quad (51)$$

then

$$\begin{aligned} Q_{\gamma_{2\varphi}}(\mathbf{z}) &= \frac{\langle \mathbf{z} || \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \frac{\langle \varphi_1 \varphi_2 || \mathbf{z} \rangle \langle \mathbf{z} | \varphi_1 \varphi_2 \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} = \gamma_{2\varphi}(\mathbf{z})_{1212} \equiv \gamma_2(\mathbf{z})_{1212} \\ &= \gamma_1(\mathbf{z})_{11} \gamma_1(\mathbf{z})_{22} - \gamma_1(\mathbf{z})_{12} \gamma_1(\mathbf{z})_{21} = \begin{vmatrix} \gamma_1(\mathbf{z})_{11} & \gamma_1(\mathbf{z})_{12} \\ \gamma_1(\mathbf{z})_{21} & \gamma_1(\mathbf{z})_{22} \end{vmatrix} \\ &= \begin{vmatrix} \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{11} & \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{12} \\ \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{21} & \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{22} \end{vmatrix} = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1}. \end{aligned} \quad (52)$$

$$\begin{aligned} \frac{\langle \varphi_1 \varphi_2 || \mathbf{z} \rangle \langle \mathbf{z} | \varphi_1 \varphi_2 \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} &= \frac{\langle \varphi_1 \varphi_2 || c(\mathbf{z}) \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | c(\mathbf{z})^\dagger \varphi_1 \varphi_2 \rangle}{\langle \varphi_1 \varphi_2 | c(\mathbf{z})^\dagger c(\mathbf{z}) \varphi_1 \varphi_2 \rangle} \\ &= \frac{|\langle \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2}{\langle \varphi_1 \varphi_2 | c(\mathbf{z})^\dagger c(\mathbf{z}) \varphi_1 \varphi_2 \rangle} \end{aligned} \quad (53)$$

Note that

$$\det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) = \det \begin{pmatrix} 1 + (\mathbf{z}^\dagger \mathbf{z})_{11} & (\mathbf{z}^\dagger \mathbf{z})_{12} \\ (\mathbf{z}^\dagger \mathbf{z})_{21} & 1 + (\mathbf{z}^\dagger \mathbf{z})_{22} \end{pmatrix}, \quad (54)$$

$$\gamma_1(\mathbf{z})_{hh'} = \left[(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1} \right]_{hh'} \quad (55)$$

and

$$\begin{aligned} 1 &= \langle \mathbf{z} || \varphi_1 \varphi_2 \rangle \langle \varphi_1 \varphi_2 | \mathbf{z} \rangle = |\langle \mathbf{z} | \varphi_1 \varphi_2 \rangle|^2 = |\langle \varphi_1 \varphi_2 | c(\mathbf{z}) \varphi_1 \varphi_2 \rangle|^2 \\ &= \left| \left\langle \varphi_1 \varphi_2 \left| \exp \left\{ \sum_{\substack{1 \leq h \leq 2 \\ 3 \leq p \leq r}} z_{ph} a_p^\dagger a_h \right\} \varphi_1 \varphi_2 \right. \right\rangle \right|^2, \end{aligned} \quad (56)$$

thus

$$Q_{\gamma_{2\varphi}}(\mathbf{z}) \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z}) = 1 \quad \forall \mathbf{z} \in \mathbb{C}^{2(r-2)}. \quad (57)$$

In this case

$$\begin{aligned}
1 &= \int Q_{\gamma_2\varphi}(\mathbf{z}) \langle \mathbf{z} | \mathbf{z} \rangle d\mu(\mathbf{z}) = \text{Tr} \{ |\varphi_1\varphi_2\rangle \langle \varphi_1\varphi_2| \} \\
&= \int d\mu(\mathbf{z}) = \int \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-(r+1)} (-2\pi i)^{-2(r-2)} d\bar{\mathbf{z}} d\mathbf{z}.
\end{aligned} \tag{58}$$

The energy functional is

$$E_N(D^2) = \int P_{K_{2N}}(\mathbf{z}) Q_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) = \int Q_{K_{2N}}(\mathbf{z}) P_{D^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}) \tag{59}$$

and the minimum value is

$$E_N(D_*^2) = K_{\varphi 1212} = \int P_{K_{22}}(\mathbf{z}) Q_{D_*^2}(\mathbf{z}) S(\bar{\mathbf{z}}, \mathbf{z}) d\mu(\mathbf{z}), \tag{60}$$

where the Q-symbol of the Reduced hamiltonian is given by

$$\begin{aligned}
Q_{K_{22}}(\bar{\mathbf{z}}, \mathbf{z}) &= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} K_{\varphi j_1 j_2 k_1 k_2} \frac{\langle \mathbf{z} | a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} \mathbf{z} \rangle}{\langle \mathbf{z} | \mathbf{z} \rangle} \\
&= \frac{1}{2} \sum_{1 \leq j_1, j_2, k_1, k_2 \leq r} \{ \eta(\delta_{j_1 k_1} \delta_{j_2 k_2} - \delta_{j_1 k_2} \delta_{j_2 k_1}) + (h_{1jk} \delta_{lm} + h_{1lm} \delta_{jk}) \} \\
&\quad \times K_{\varphi j_1 j_2 k_1 k_2} \left\{ \gamma_1(\mathbf{z})_{j_1 k_1} \gamma_1(\mathbf{z})_{j_2 k_2} - \gamma_1(\mathbf{z})_{j_1 k_2} \gamma_1(\mathbf{z})_{j_2 k_1} \right\},
\end{aligned} \tag{61}$$

and its P-symbol $P_{K_{22}}(\mathbf{z})$ has to be found via the inverse Berezin transform

$$P_{K_{22}}(\mathbf{z}) = [\mathfrak{B}\mathfrak{e}^{-1}(Q_{K_{22}}(\mathbf{z}))](\mathbf{z}) = \int \mathfrak{B}\mathfrak{e}^{-1}(\bar{\mathbf{z}}, \mathbf{z}') Q_{K_{22}}(\mathbf{z}') d\mu(\mathbf{z}'), \tag{62}$$

and

$$\begin{aligned}
Q_{D_*^2}(\mathbf{z}) &= [D_{*2}]_{1212} \gamma_2(\mathbf{z})_{1212} \\
&= [D_{*2}]_{1212} \{ \gamma_1(\mathbf{z})_{11} \gamma_1(\mathbf{z})_{22} - \gamma_1(\mathbf{z})_{12} \gamma_1(\mathbf{z})_{21} \} \\
&= \gamma_1(\mathbf{z})_{11} \gamma_1(\mathbf{z})_{22} - \gamma_1(\mathbf{z})_{12} \gamma_1(\mathbf{z})_{21} = \det(\mathbb{I}_2 + \mathbf{z}^\dagger \mathbf{z})^{-1}.
\end{aligned} \tag{63}$$

The energy is given as

$$\begin{aligned}
E(D_*^2) &= \inf_{D^2 \in S_{22} = S_2} \text{Tr} \{ K_{22} D^2 \} \\
&\parallel \\
E(Q_{D_*^2}) &= \inf_{Q \in \mathfrak{Q}_{12}} \left\{ \int P_{K_{22}}(\mathbf{z}) d\mu(\mathbf{z}) \right\} \\
&= K_{\varphi 1212},
\end{aligned} \tag{64}$$

where

$$K_{22} = \int P_{K_{22}\varphi}(\mathbf{z}) |c(\mathbf{z}) \varphi_1 \varphi_2\rangle \langle c(\mathbf{z}) \varphi_1 \varphi_2| d\mu(\mathbf{z}). \tag{65}$$

C. BiOrthogonal Basis

$$A|\varphi\rangle = |\varphi\rangle \mathbb{A} \quad (66)$$

$$\langle\psi|\varphi\rangle = \mathbb{I} \quad (67)$$

$$\Delta_\psi = \langle\psi|\psi\rangle \Rightarrow \langle\psi|\psi\rangle \Delta_\psi^{-1} = \langle\psi|\Delta_\psi^{-1}\psi\rangle = \mathbb{I} \Rightarrow |\varphi\rangle = |\Delta_\psi^{-1}\psi\rangle \quad (68)$$

$$P = |\varphi\rangle \Delta_\varphi^{-1} \langle\varphi| = |\psi\rangle \Delta_\psi^{-1} \langle\psi| = |\varphi\rangle \langle\psi| = |\varphi\rangle \langle\psi| \quad (69)$$

II. EXTRA

$$\sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} \mathcal{P}_{p_1 p_2 q_1 q_2}(\mathbf{z}') \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1} \quad (70)$$

$$\begin{aligned} &= \int \Upsilon_{j_1 j_2 k_1 k_2}(\mathbf{z}') |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \\ \Upsilon_{j_1 j_2 k_1 k_2}(\mathbf{z}') &= \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} \mathcal{P}_{p_1 p_2 q_1 q_2}(\mathbf{z}') \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1}. \end{aligned} \quad (71)$$

this gives

$$\Upsilon_{j_1 j_2 k_1 k_2}(\mathbf{z}') = \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} Q_{p_1 p_2 q_1 q_2}(\mathbf{z}) \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-2} \quad (72)$$

and

$$\mathcal{P}_{j_1 j_2 k_1 k_2}(\mathbf{z}) = \int \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} Q_{p_1 p_2 q_1 q_2}(\mathbf{z}) \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-2} |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}'). \quad (73)$$

As

$$Q_A(\mathbf{z}) = \int \mathcal{P}_A(\mathbf{z}') |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \quad (74)$$

one obtains

$$\begin{aligned} \mathcal{P}_A(\mathbf{z}) &= \int \sum_{1 \leq p_1 < p_2, q_1 < q_2 \leq r} A_{j_1 j_2 k_1 k_2} \mathcal{P}_{p_1 p_2 q_1 q_2}(\mathbf{z}') \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1} |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \\ &= \int \Upsilon_A(\mathbf{z}') |\langle \mathbf{z} | \mathbf{z}' \rangle|^2 d\mu(\mathbf{z}') \\ \Upsilon_A(\mathbf{z}') &= \sum_{\substack{1 \leq p_1 < p_2, q_1 < q_2 \leq r \\ 1 \leq j_1 < j_2, k_1 < k_2 \leq r}} A_{j_1 j_2 k_1 k_2} \mathcal{P}_{p_1 p_2 q_1 q_2}(\mathbf{z}') \Delta_{p_1 p_2 q_1 q_2 j_1 j_2 k_1 k_2}^{-1} \end{aligned} \quad (75)$$

As

$$\mathcal{P}_{p_1 p_2 q_1 q_2}(\mathbf{z}) = \sum_{1 \leq j_1 < j_2, k_1 < k_2 \leq r} Q_{j_1 j_2 k_1 k_2}(\mathbf{z}) \Delta_{j_1 j_2 k_1 k_2 p_1 p_2 q_1 q_2}^{-1}$$

A.